

Week 4

Data Heterogeneity and Advanced Federated Optimization

Why identical algorithms behave differently across clients

Graduate course · English instruction



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The Problem with “One Distribution”

In federated learning, each client may see a different data distribution, amount of data, and local objective.

Statistical heterogeneity

Local data distributions differ across clients.

Optimization consequence

Local gradients and local minimizers can point in different directions.

System consequence

Clients may also differ in speed, local epochs, and availability.

same global model



different local objectives



drift + slower convergence

Heterogeneity is not an implementation detail; it changes the optimization problem itself.

Learning Objectives

Diagnose heterogeneity

Distinguish label skew, feature skew, quantity skew, and concept shift.

Explain client drift

Connect non-IID local objectives to divergent local updates.

Understand correction

Explain the key ideas behind FedProx, SCAFFOLD, and FedNova.

Compare algorithms

Match each correction mechanism to the problem it addresses.

By the end, you should be able to look at a non-IID setting and predict which baseline is likely to fail—and why.

IID vs. Non-IID Client Data

IID idealization

Each client mini-batch behaves like a sample from the same underlying global distribution.



IID: similar local objectives

Non-IID reality

Clients collect data locally, so distributions, labels, features, preferences, and sample sizes differ.



Non-IID: different local objectives

Federated optimization must work despite distribution mismatch across clients.

Common Forms of Statistical Heterogeneity

Label distribution skew

Clients observe different class proportions.

Feature distribution skew

The same label can look different across devices, locations, hospitals, or sensors.

Concept shift

The mapping from features to labels differs across clients.

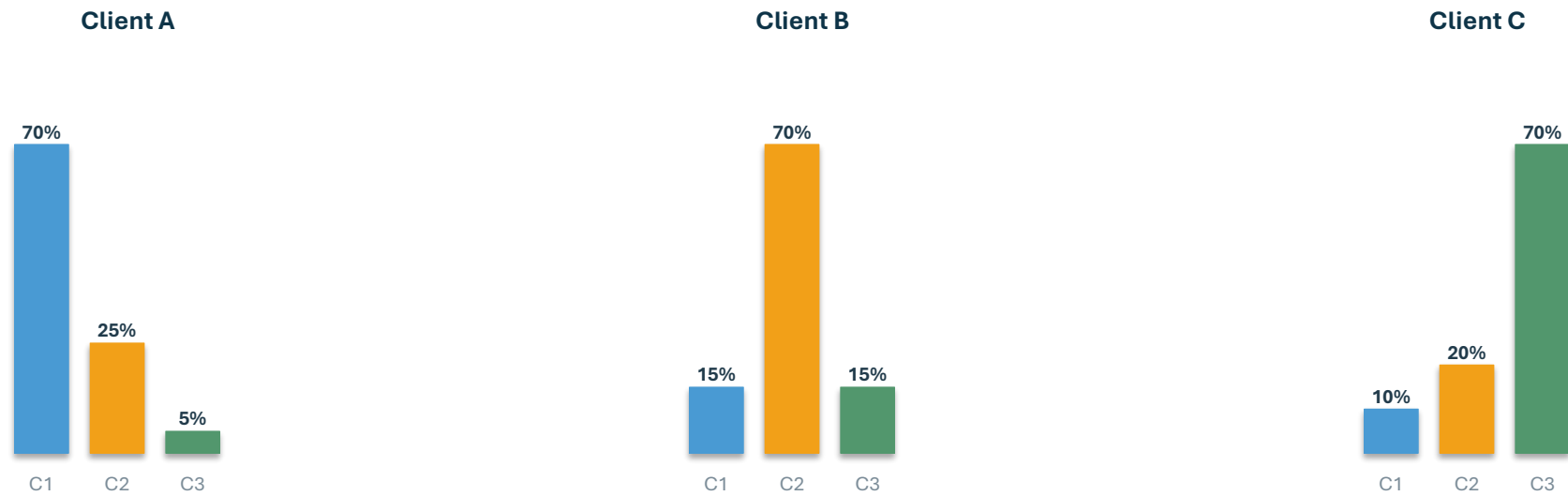
Quantity skew

Clients hold very different numbers of samples.

Non-IID is an umbrella term—not a single data pathology.

Non-IID Data Is Not One Single Phenomenon

Example: label distributions across three clients



Same task, different local class balance → different local gradients.

One label-skew example does not cover feature skew, concept shift, or quantity imbalance.

Local Objectives Can Disagree

$$F_k(\theta) = \mathbb{E}_{z \sim P_k}[\ell(\theta; z)] \quad \text{while} \quad F(\theta) = \sum_k p_k F_k(\theta)$$

When P_k differs across clients, the minimizer and gradient of each local objective can differ from the global objective.

Gradient disagreement

∇F_k may point in conflicting directions.

Local optima

θ_k^* may be far apart.

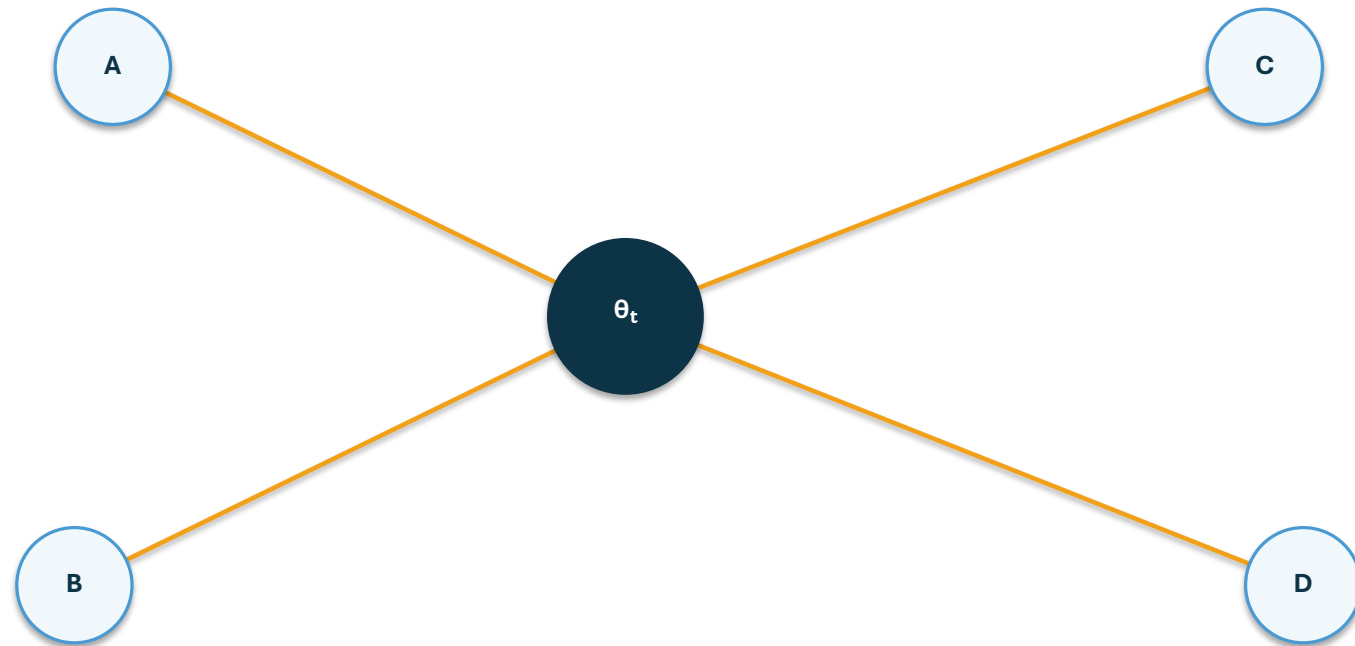
FedAvg effect

Averaging local solutions can deviate from the desired global descent direction.

This mismatch is the mathematical source of client drift.

Client Drift Under Heterogeneous Data

Client drift grows when local objectives pull in different directions.



More local steps $E \rightarrow$ larger separation among client models

After many local updates, simple averaging may no longer approximate one clean global descent step.

FedProx: Keep Local Models Close to the Global Model

$$\text{minimize locally: } F_k(\theta) + (\mu/2) \|\theta - \theta_t\|^2$$

The proximal penalty discourages a client from moving too far away from the current global model.

Control

μ controls how strongly the proximal term constrains local updates.

Benefit

More stable local optimization under heterogeneous data.

Trade-off

Too much regularization can limit useful local adaptation.

FedProx changes the local objective—not the server aggregation rule.

SCAFFOLD: Correct Client Drift

Use control variates to estimate and correct the mismatch between local update directions and the global direction.

Client control c_k

Tracks local drift information and corrects local updates.

Server control c

Represents a global correction direction.

Goal

Reduce variance caused by heterogeneous client gradients.

raw local gradient



+ control correction



less drift

SCAFFOLD targets direction bias rather than constraining distance.

FedNova: Normalize Heterogeneous Local Work

Normalize accumulated local updates so clients performing different numbers of local steps contribute on a comparable scale.

Problem

Different numbers of local steps create objective inconsistency.

Correction

Normalize the effective local update by the amount of local work.

Result

Server aggregation better reflects a consistent optimization direction.



2 steps



5 steps



10 steps

Normalization removes the unfair influence of unequal local computation.

FedAvg, FedProx, SCAFFOLD, and FedNova

Method	Main issue	Core idea	Extra state / tuning
FedAvg	Baseline heterogeneity	Average local models	Low
FedProx	Local drift / unstable local solve	Proximal regularization	μ
SCAFFOLD	Gradient drift	Control variates	Control states
FedNova	Unequal local work	Normalize local updates	Normalization metadata

The methods solve different failure modes—do not treat them as interchangeable.

Evaluation Under Heterogeneity

Global accuracy

How well does one global model perform overall?

Client mean

Average performance across clients.

Worst-client / tail

Do some clients systematically suffer?

Variance across clients

How unequal are client outcomes?

Convergence speed

Rounds needed to reach target performance.

Communication cost

Accuracy must be interpreted together with system cost.

A method can improve mean accuracy while worsening the worst clients—or vice versa.

Experimental Design for Non-IID Studies

Control heterogeneity

Vary one factor at a time: label skew, feature skew, quantity skew, or local steps.

Hold budgets fixed

Compare methods under the same rounds, clients per round, and local computation.

Report distribution

Show client-level metrics, not only one global average.

Use multiple seeds

Optimization and client sampling can introduce substantial variance.

Fair comparisons require controlling both statistical heterogeneity and system budgets.

In-Class Case Analysis

Scenario

Four clinics have different disease prevalence and different numbers of local SGD steps.

Question 1

Which heterogeneity is statistical? Which is system-induced?

Question 2

When would FedProx help? Which issue does FedNova address better?

Question 3

What metrics would reveal whether one clinic is harmed?

Try to map every symptom to a mechanism before choosing an algorithm.

Key Takeaways

- 1 **Non-IID data includes several distinct distribution shifts; one label-skew example is not enough.**
- 2 Heterogeneous local objectives create client drift and can bias simple FedAvg updates.
- 3 **FedProx regularizes local movement; SCAFFOLD corrects drift; FedNova normalizes unequal local work.**
- 4 Algorithm comparisons should control computation and communication budgets and report client-level outcomes.
- 5 Next week: system heterogeneity, client selection, and communication efficiency.

Mental model: identify the heterogeneity → identify the optimization failure mode → choose the correction.

Reading & Preparation

Main textbook

Ludwig & Baracaldo (2022)
Review data heterogeneity and federated optimization chapters.

Supplementary references

FedProx, SCAFFOLD, and FedNova foundational papers; revisit Kairouz et al. (2021).

Prepare for Week 5 — System Heterogeneity, Client Selection, and Communication Efficiency

- 1 List three types of system heterogeneity.
- 2 Think about stragglers and partial participation.
- 3 Review why communication can dominate FL training time.

Bring one example where system constraints—not data distribution—cause the main FL bottleneck.

Q & A

Thank You.

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